

Data Transformation and Feature Engineering

AWS preparation tools, preprocessing, features, embeddings, RAG, and privacy



Transformation connects raw data to model-ready input

01

02

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Inspect

Understand schema, ranges, categories, missing values, and sensitive fields.

Clean

Correct invalid data, duplicates, inconsistent formats, and missing values.

Transform

Scale, encode, split, combine, or reshape values.

Represent

Create features, tokens, embeddings, chunks, and metadata.

Validate

Confirm quality, consistency, privacy, and suitability for the model.



AWS transformation tools serve different workloads

AWS Glue		Managed data integration and ETL	Discovers, catalogs, transforms, and moves data across sources.
Glue DataBrew		Visual data preparation	Profiles and transforms data through a visual interface.
Amazon EMR		Large-scale Spark processing	Supports flexible distributed data processing frameworks.
Data Wrangler		SageMaker data preparation	Connects, visualizes, transforms, and exports ML data flows.



Common preprocessing methods

Scaling



Change the numeric range

Prevents larger numbers from dominating smaller numbers.

Standardization



Center values around the mean

Uses distance from the mean to create a common scale.

Encoding



Represent categories numerically

Creates inputs that a model can process.

Binning or logs



Group values or reduce skew

Makes difficult distributions easier to model.



Feature engineering adds useful meaning

Traditional ML features

Examples include ratios, time intervals, grouped totals, interaction terms, normalized values, and domain-specific indicators.

Feature management

SageMaker Feature Store can organize, share, and serve features while helping training and inference use consistent definitions.



Embeddings turn meaning into numerical space

What an embedding represents

An embedding converts text, images, or other content into a vector. Similar items appear closer together according to the model.

Why embeddings matter

Vectors support semantic search, recommendations, clustering, retrieval, and context selection for generative AI.



RAG preparation shapes what a model can retrieve

01 Parsing

Extract usable text and structure from source documents.

02 Chunking

Divide documents into sections that preserve enough context without becoming too large.

03 Metadata

Attach source, date, topic, access level, and other fields for filtering and traceability.

04 Protection

Mask, redact, or anonymize sensitive information before wider processing or retrieval.



Preparation must stay consistent from training to inference

Choose the tool that fits the scale and workflow.
Apply transformations for a clear statistical reason.
Manage features and embeddings as reusable data products.
Protect sensitive information throughout preparation.

NEXT

**Part 3: Data quality,
labeling, and bias**

